

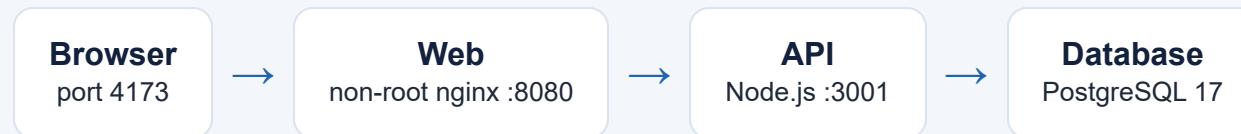
ARCHITECTURE OVERVIEW

Full-Stack CI/CD Pipeline

React + Node.js API + PostgreSQL, tested as one system and released as immutable web and API containers.

APPLICATION

Runtime architecture



Single public origin

nginx serves React and proxies `/api/*`. API and PostgreSQL stay private.

Persistent data

PostgreSQL uses a named volume. Application rollback never deletes data.

Reusable workflow layout

```
ci.yml cd.yml
├ reusable-test.yml │ reusable-deploy.yml
├ reusable-security.yml └ reusable-rollback.yml
└ reusable-docker.yml
```

Reusable workflows are flat under `.github/workflows`, as required by GitHub Actions.

CONTINUOUS INTEGRATION

Test the complete system

Web

- ESLint + Prettier
- TypeScript
- Vitest coverage

API + database

- API unit tests
- PostgreSQL service
- Migration + integration test

Full-stack E2E

- Docker Compose
- Real contact API
- Five Playwright browsers

SECURITY

Gates across both applications

Dependencies

Separate web and API lockfile audits; reviewed dependency upgrades only.

Source

TruffleHog, CodeQL, and Trivy filesystem SARIF.

Containers

Web and API SBOMs, provenance attestations, blocking HIGH/CRITICAL scans.

RELEASE

Build once, publish by CalVer

```
ghcr.io/<owner>/platform-web:YYYY.MM.N  
ghcr.io/<owner>/platform-api:YYYY.MM.N
```

Traceable

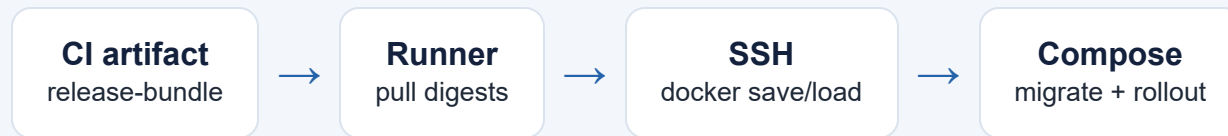
:staging, :sha-short, SBOM, git revision, and provenance.

Immutable deployment

CD deploys `image@sha256:digest`, never a mutable tag.

CONTINUOUS DELIVERY

Staging deployment



The VM does not need GHCR access. Tailscale is optional; host keys remain pinned.

RESILIENCE

Health and rollback

Release validation

- PostgreSQL health
- API readiness
- web health
- external smoke routes

Failure path

- restore prior web/API digests
- preserve PostgreSQL volume
- do not reverse migrations
- create deployment issue

FEEDBACK

Issues, wiki, and artifacts

Failure issues

Full-stack tests, security, container release, migration, Compose deploy, and rollback.

Wiki reports

Run metadata, stage results, job links, remediation, and release artifacts.

Artifacts

web-dist, coverage, Playwright report, release metadata, Compose, scripts, SBOMs.

SUMMARY

One pipeline. One full stack.

PR validation → secure immutable release → Compose staging deployment → automatic application rollback.